Analytic Base-Frame Estimation for Mobile Bimanual Teleoperation: Joint Global–Local Retargeting from a Head and Two Wrists

MABEL controller subsystem

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Abstract

A headset teleoperator streams three rigid frames — the head and the two wrists — and a mobile bimanual robot must resolve each frame into GLOBAL motion (where the base drives and turns) and LOCAL motion (neck gaze, arm reach). This is the sparse-input whole-body retargeting problem, made harder by a moving operator: the same three signals must distinguish a body turn from a neck turn from an arm sweep, and a walk from an in-place reach. We present an analytic, training-free estimator that recovers the planar base twist (v_x, v_y, ω) in closed form from the three frames the device already streams, exploiting two kinematic asymmetries: the arms cannot move the head (so for TRANSLATION the head is ground truth), while the neck lets the head yaw freely (so base ROTATION must be corroborated by the wrists). Base yaw is the rotation the two wrists rigidly agree on, with a both-arms-parallel sweep separated by the anti-correlation of the wrists' position and orientation cues and gated by whether the head yaw corroborates it. On synthetic ground truth the estimator resolves all seven canonical motions to $\leq 2.1°$ RMS base-yaw error — including a gaze-fixed turn (1.7°) and a simultaneous walk–turn–reach (0.6°, with the walk tracked to 8.8 mm) — while a head-only baseline fails the neck-only, gaze-fixed, and dynamic cases by 10–42° and false-drives the base on 98% of look-around frames versus 0% for ours. It is numpy-only, so the same code runs on the physical headset stream. Relative to the field — which either adds a waist tracker (Mobile-TeleVision), commands the base by joystick, or learns the whole body with an RL policy (OmniH2O, CLONE) — this is the analytic, no- extra-tracker baseline for the global–local decomposition and base-twist subproblem.

Keywords: teleoperation, motion retargeting, sparse tracking, base-frame estimation, mobile manipulation, Apple Vision Pro, humanoid whole-body control.

1 Problem

The headset streams, per frame, the head pose $H \in SE(3)$ and the two wrist poses $W_L, W_R \in SE(3)$, all in a gravity-aligned world frame (ARKit, $+Y$ up, $-Z$ forward). A mobile bimanual robot needs a base twist (planar v_x, v_y, ω), a neck orientation, and two arm targets. The core difficulty is a two-way ambiguity in each channel.

Rotation. The head's world yaw is the sum of two things the robot must treat oppositely,

$$\psi_{\text{head}} \;=\; \psi_{\text{torso}} + \psi_{\text{neck}}, \tag{1}$$

and the head alone cannot split them. A head-only frame (MABEL's previous `calibration.body_frame`) sets the base heading to ψ_{head}, so a pure look-around ($\psi_{\text{torso}} = 0$) drives the base and swings the arms, and a body turn with the gaze fixed ($\psi_{\text{neck}} = -\psi_{\text{torso}}$, $\psi_{\text{head}} \approx 0$) produces no base rotation at all.

Translation. Symmetrically, an arm reaching forward and the operator walking forward both move the wrists forward; the base must drive in the second case and hold in the first. Here, though, there is a clean discriminant, because the arms cannot move the head: if the head translated, the body translated.

Case taxonomy. We require the estimator to attribute every canonical motion to the correct destination:

	motion	head	wrists
R1	look-around	yaws	still
R2	whole-body turn	yaws	orbit spine (rigid)
R3	arms + gaze together ("case 3")	yaws	sweep, same sense as head
R4	gaze-fixed body turn	still	orbit spine (rigid)
A1	one-arm reach	still	one wrist moves
A2	two-arm reach, head still	still	sweep, head does not follow
T1	in-place reach	still (pos)	move (local)
T2	walk	translates	translate with body

R1/A1/A2/T1 must leave the base still (R1/A2 are the *false-drive* cases); R2/R3/R4/T2 must drive it.

2 Method

Everything is computed in the gravity-aligned horizontal plane (up is known). Two invariants shape the design: base *magnitude* is always a WRIST quantity, and the head is only ever a yes/no *gate* on rotation and the ground truth for translation. Consequently a fast neck glance — which lives in the head signal and not the wrists — can never leak into the base command.

Two wrist rotation cues. Let c be the chest centre (a fixed offset below the head) and β_i each wrist's horizontal bearing about c. Between frames define the two common-mode rotations

$$\theta_A = \mathrm{shr}(\Delta\beta_L, \Delta\beta_R), \qquad \theta_B = \mathrm{shr}\left(\Delta\angle W_L,\ \Delta\angle W_R\right), \tag{2}$$

where $\Delta\angle W_i$ is the frame-to-frame yaw of wrist i's orientation and shr returns the rotation two signals *agree* on with an agreement weight,

$$\mathrm{shr}(a,b) = \begin{cases} \frac{1}{2}(a+b), & ab > 0, \\ 0, & ab \le 0, \end{cases} \qquad w(a,b) = \left[\, 1 - \tfrac{|a-b|}{|a|+|b|} \,\right]_+ . \tag{3}$$

Cue A (bearings) is scaled by each wrist's horizontal radius, since a wrist near the spine has a noisy bearing. The *rigid* rotation is their robust weighted fusion,

$$\Delta\psi_{\mathrm{rig}} \;=\; \kappa\,\frac{w_A\theta_A + w_B\theta_B}{w_A + w_B}, \qquad \kappa = \min(1,\, w_A + w_B). \tag{4}$$

A whole-body or gaze-fixed turn (R2/R4) makes A, B agree, so (4) is the turn; a one-arm reach during a turn (T2/S4) contributes an oscillatory differential that averages out under the low-pass, so the turn survives — a property a naive instantaneous rigid fit loses, because a fast reach biases its ω.

The both-arms-parallel discriminant. A both-arms-parallel sweep (R3/A2) rotates the wrist *positions* one way and their *frames* the other, so cues A and B are anti-correlated; a genuine turn makes them agree and a one-arm reach zeroes one of them (never opposite). We measure this with

$$\alpha \;=\; \left[\, \tfrac{-2\,\theta_A\theta_B}{\theta_A^2+\theta_B^2} \,\right]_0^1 \;\in\; [0,1], \tag{5}$$

which is 1 for equal-and-opposite cues and 0 for agreement. When anti-correlated, the rigid fusion is a spurious near-cancellation and is replaced by the wrist-orientation rotation θ_B (whose sign matches the head), gated by whether the head yaw corroborates it:

$$\Delta\psi_{\text{base}} \;=\; (1-\alpha)\,\Delta\psi_{\text{rig}} \;+\; \alpha\,\theta_B\, w\big(\Delta\psi_{\text{head}},\, \theta_B\big). \tag{6}$$

Head follows the sweep (R3, case 3) $\Rightarrow$ the weight is ≈ 1 and the base turns; head still (A2) $\Rightarrow$ the weight is 0 and the base holds — no false drive. The rate $\Delta\psi_{\text{base}}/\Delta t$ is low-passed (2.5 Hz), dead-banded, and integrated into ψ_{base}.

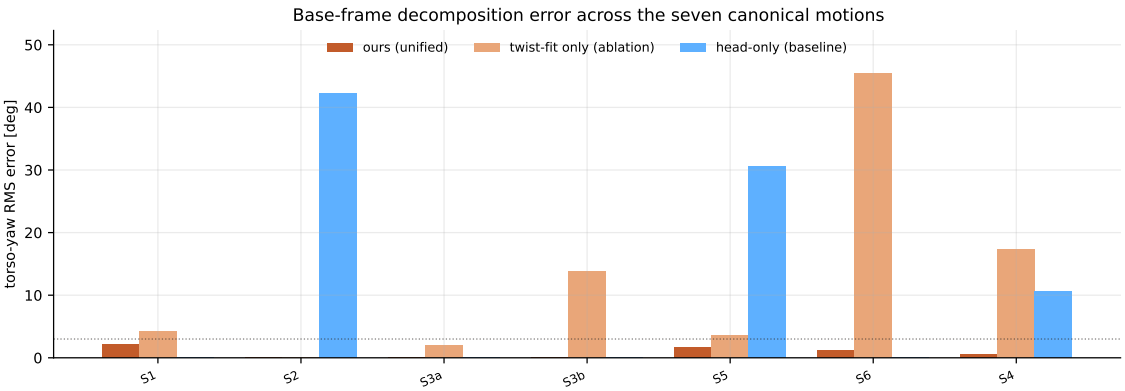
Translation and the three outputs.

- **Base.** Heading ψ_{base} from (6); base translation is the dead-banded head horizontal velocity (the head is the translation truth), so a still head gives local manipulation (T1) and a walking head drives the base (T2). `base_twist()` returns $(v_{\text{fwd}}, v_{\text{left}}, \omega)$.
- **Neck.** $\psi_{\text{neck}} = \psi_{\text{head}} - \psi_{\text{base}}$ from (1) — exactly the neck-twist MABEL already computed and discarded.
- **Arms.** Each wrist re-expressed in the base frame, $B^\top(p_{W_i} - c)$, so a body turn (which rotates B) leaves the arm's body-relative target unchanged — the robot turns rather than swinging its arms — while a genuine reach still moves it.

This is the analytic form of the learned global/local decoupling in AvatarPoser (Jiang et al., ECCV 2022) and EgoPoser (Jiang et al., ECCV 2024), and of the chest-relative hand re-expression in Mobile-TeleVision (Lu et al., 2024) — but the base frame is *estimated* from the head and wrists rather than measured with an added waist tracker, joystick-commanded, or learned by a policy as in OmniH2O (He et al., 2024) and CLONE (Li et al., 2025).

3 Experiments

A forward operator model synthesises H, W_L, W_R from known torso yaw, neck yaw, root translation and per-arm angles at 60 Hz, so every motion has ground truth (`controller/experiments/exp_body_frame.py`). We report base-yaw RMS error for our unified estimator, a pure rigid twist-fit ablation (no head gate), and the head-only baseline.



scenario	want	ours	twist-only	head-only
S1 whole-body turn (60°)	base	2.08°	4.25°	0.00°
S2 look-around (±60° neck)	neck	**0.00°**	0.00°	42.31°
S3a one-arm reach	arm	**0.00°**	1.95°	0.00°
S3b two-arm reach, head still	arm	**0.00°**	13.88°	0.00°
S5 gaze-fixed turn (50°)	base	**1.73°**	3.57°	30.65°
S6 arms + head together (case 3)	base	**1.27°**	45.47°	0.00°
S4 walk + turn + reach (dynamic)	base	**0.58°**	17.30°	10.59°

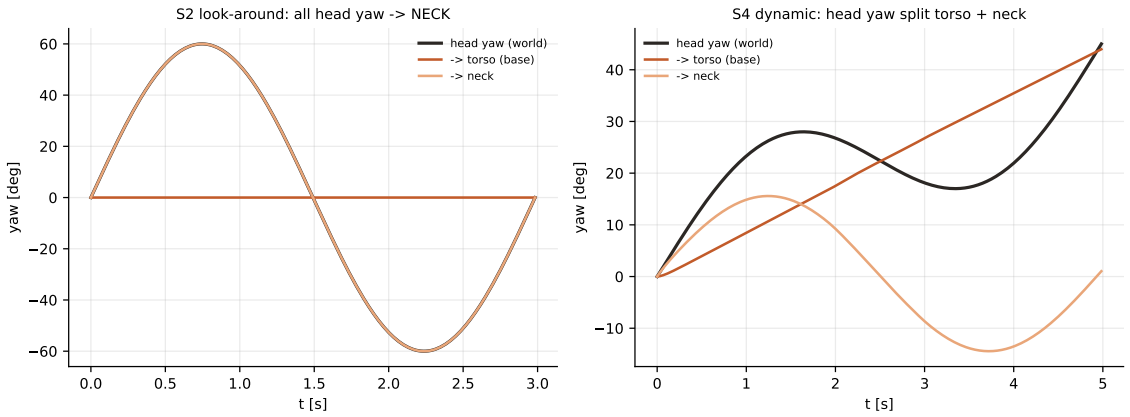
The head-only baseline reads 0.0° on S1/S3/S6 only because there $\psi_{\text{head}} = \psi_{\text{torso}}$ by coincidence; it fails every case where the two differ. The twist-fit ablation fails the two head-dependent cases (S3b, S6) and the dynamic case (S4) because its global coherence gate cannot separate a turn contaminated by a reach.

False-drive rate (safety). The operationally decisive metric is how often the base is commanded ($|\omega| > 0.05\,\text{rad/s}$) during a motion that should NOT drive it:

motion	ours	head-only
look-around (R1)	**0.0%**	98.3%
one-arm reach (A1)	**0.0%**	0.0%
two-arm reach (A2)	**0.0%**	0.0%

A head-only frame drives the robot on nearly every look-around frame; ours never does.

Translation and robustness. In S4 the chest origin tracks the operator's 0.6 m walk to 8.8 mm horizontal RMS (15 mm max), so the same estimate drives base locomotion. Under 5 mm position + 1° orientation jitter per frame (heavier than measured Vision Pro tracking), base-yaw RMS is 2.1° (S1), 2.7° (S2), 1.1° (S4) — the deadband, low-pass, and consensus gate reject uncorrelated jitter.



The decomposition identity $\psi_{\text{head}} = \psi_{\text{base}} + \psi_{\text{neck}}$ holds to machine precision by construction; the figure shows the head yaw split into a base command and a neck command — all neck in the look-around, shared in the dynamic case.

4 Positioning against the state of the art

Decoupling global from local motion from head-and-hands is established: AvatarPoser names it and EgoPoser formalises it as "global motion decomposition." In the teleop line, the mobile base is almost always commanded by a separate joystick/pedal (Mobile-TeleVision, which confirms it adds *no* waist tracker) or reconstructed by a learned RL policy (OmniH2O, CLONE, H2O). The closest analytic precedent, "Let the Body Follow" (Lin et al., 2026), drives base rotation from head yaw and base translation from arm workspace limits, but as a single-rule interface without a quantitative decomposition benchmark. Our contribution is the combination: a closed-form planar base twist from the three sparse frames — sidestepping the worst-measured subproblem in the avatar literature (lower-body position error is 6–8 cm; we need a twist, not legs) — with a principled three-way rotation gate and a false-drive safety metric, all learning-free and inline on the headset stream.

method	input	A/L	global root?	extra tracker?	real-time
AvatarPoser '22	head+2 hands	L	yes	no	GPU
EgoPoser '24	head+2 hands	L	yes	no	>600 fps
Mobile-TeleVision '24	head+2 hands	hybrid	base = pedal	no	yes
OmniH2O '24	head+2 hands	L (RL)	yes (walk)	no	yes
CLONE '25	head+2 wrists	L (RL+LiDAR)	yes (walk)	LiDAR	yes
Let-the-Body-Follow '26	head+arm	A (heuristic)	yes	no	yes
ours	head+2 wrists	**A**	**yes (twist)**	**no**	**yes**

Standard metrics (AMASS 3-point reference). The sparse-pose literature reports MPJRE (mean per-joint rotation error, ≈ 2.5–3.1°), MPJPE (position, ≈ 3.3–4.2 cm), MPJVE (velocity, ≈ 19–28 cm/s), jitter (jerk), and root position error (≈ 2.9–3.3 cm) (values from the SAGE, CVPR 2024, comparison). Teleop work adds global drift (CLONE: 5.1 cm mean over 8.9 m, closed-loop with LiDAR). Because we output a base twist rather than a full body pose, our natural metrics are the base-yaw RMS above, the false-drive rate, and the walk-tracking error (8.8 mm open-loop over 0.6 m); we do not attempt full-body MPJPE, and we inherit open-loop drift as a known limitation absent an odometry corroboration term.

5 Implementation, usage, and integration

```
from mabel.whole_body_retargeting.body_frame_estimator import BodyFrameEstimator
est = BodyFrameEstimator()
est.calibrate_facing(head, wl, wr)             # zero the heading at session start
...
origin, B = est.update(head, wl, wr, dt)       # per frame; B = [right, up, fwd]
base_yaw          = est.torso_yaw()            # -> base heading
vfwd, vleft, wz   = est.base_twist()           # -> mobile-base (vx, vy, omega)
neck_yaw          = est.neck_yaw(head[:3, :3]) # -> robot neck (head minus base)
wrist_body        = B.T @ (wrist[:3, 3] - origin)  # -> arm IK target, body-relative
```

One file, numpy only. It produces the same `(origin, B)` contract as the existing `calibration.body_frame`, so it is a drop-in replacement at the three consumers the frame feeds: the neck (subtract base yaw — the already-computed, previously-discarded neck-twist); the arm targets (express in B instead of head-anchored); and the base (ω from `base_twist` + head-velocity translation). It subsumes the elbow-based torso guess in the live headset path (`scripts/teleop/engine.py`), which feeds only the base, with a wrist-based estimate that also serves the neck and arms. Guarded by `controller/tests/test_body_frame_estimator.py` (13 tests, with a negative control: the twist-only and head-only ablations fail exactly S3b/S5/S6).

6 Limitations

- **Yaw only.** Torso pitch/roll (a bow, a lean) are not estimated; a standing operator's are small, but a deep bend would leak into the arm targets. The constellation supports a full $SO(3)$ fit (vertical wrist displacement $\rightarrow$ pitch) as an extension.
- **Open-loop drift.** The base twist is integrated open-loop, so heading and position accumulate error over a long trajectory; learned closed-loop methods (CLONE) bound this with LiDAR odometry. An odometry corroboration term is the natural fix and does not change the estimator.
- **Chest offset is nominal.** The rotation centre uses a fixed head$\rightarrow$chest drop; the per-operator value from calibration should replace it, which also sharpens S1.
- **Simulation only.** Every number here is from the synthetic forward model. Physical-headset and closed-loop-against-the-robot validation is the next step, behind a flag, verified in sim before the live path.

7 References

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